

sent mmWave technology, and early versions of 28GHz equipment. Test equipment users need to ensure they have flexible test equipment to both validate their designs now and for the longer term.

Infrastructure developments to support 5G rollout

5G base stations use high frequency bands and have small coverage, leading to the challenge of limited space for multiple sites. Sharing infrastructure and power supply resources with energy suppliers, tower providers and carriers provide physical bases to deploy 5G base stations. Using 5G antenna technology, tower providers can mount multiple types of antennae on one mono-pole tower.

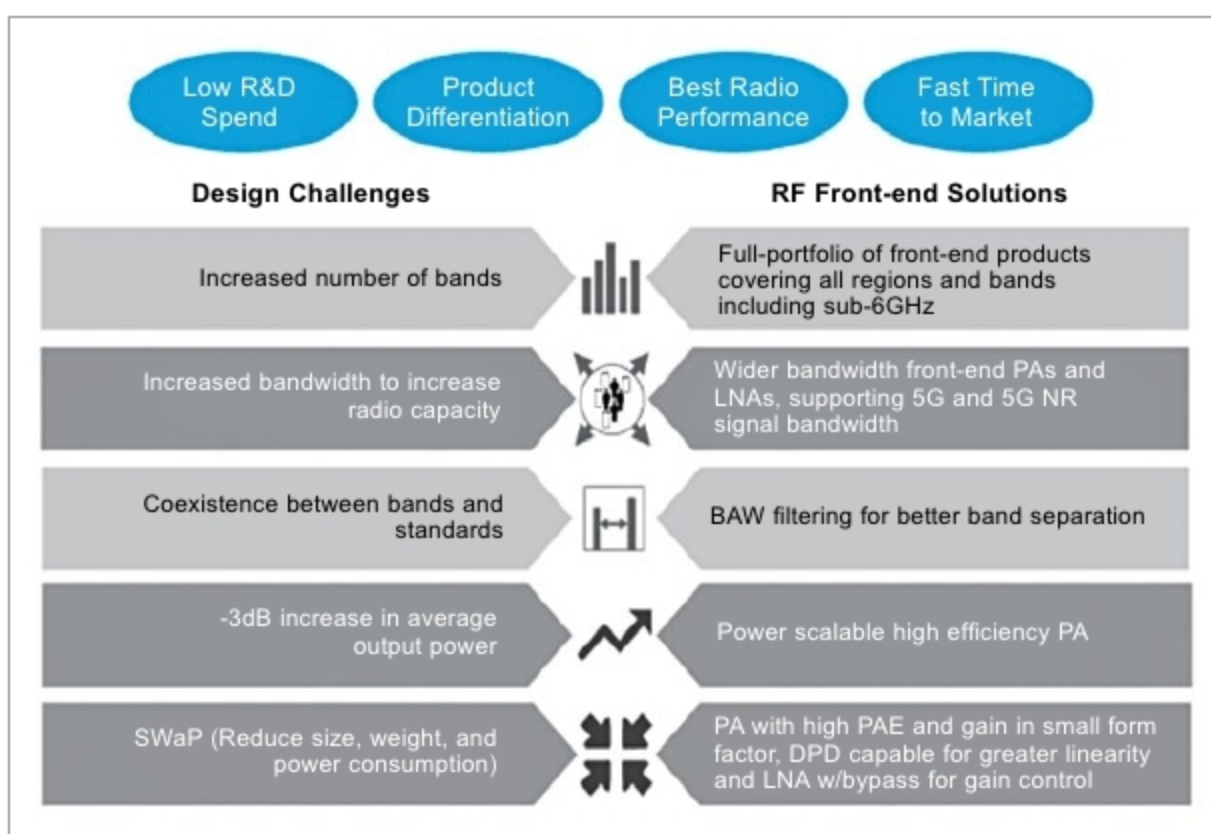
The intent of small cells is to diminish the load on macro cells by augmenting network capacity and provide extended coverage. With the market transitioning to 5G, a whole new spectrum is opening with bandwidth of more than 100MHz available within 2.6GHz to 5GHz range. This additional spectrum will support high data rates and new market opportunities.

Further, higher order multiple-input-multiple-output (MIMO) is the standard macro architecture enabling higher throughput and data rates.

Traditionally, small cells have been deployed in a 2-transmit/2-receive (2T/2R) MIMO configuration, but with 5G, that architecture will be expanded to 4T/4R for increased throughput. Advanced antenna array architecture and beamforming further help reduce co-cell interference.

By shifting more frequency translation and filtering from analogue to the digital domain, devices like AD9081/2 provide designers with software configurability to customise their radios. The multi-channel MxFE platform meets the needs of other wide-bandwidth applications in 5G (T&M) equipment, broadband cable video streaming, multi-antenna phased-array radar systems and low-earth-orbit satellite networks.

As frequencies increase, so does the amount of electromagnetic interference. To tackle this problem, Heraeus Printed Electronics has developed a silver ink printer that ensures



Small-cell manufacturer challenges and solutions (Credit: www.qorvo.com)

proper functioning of high-frequency onboard chips and their ultra-fast data transmission. The system consists of a special silver ink formula and a manufacturing machine that can apply the shielding. In contrast to sputtering, the protective silver layer is applied to the carrier object with sub-micrometre precision by using printing nozzles. This not only saves material and cost but also leads to much better shielding performance.

Advanced testing and measurement technologies

Measurement results of a device under test (DUT) should not be setup-dependent, which requires calibration for each measurement setup. OTA link adds a significant amount of attenuation to the measurement setup due to free space path loss, in addition to losses from cables and other connected components.

Stable measurement distance and perfect mechanical alignment are crucial for OTA measurement accuracy to avoid attenuation through positioning errors. Calibration information of the OTA test setup is stored either on the measurement instrument or some external software controlling the measurement setup.

There are a few major players in the development of 5G testing services for a 5G-enabled connected world. Some of their advanced testing services are given below.

Testing of location-based service (LBS). As services become more personalised, location determination technologies will play a critical role. Along with introduction of new satellite location technologies in cellphones such as modernised GPS (L5) and GALILEO, it is critical that LBS performance of mobile devices is verified extensively before commercial launch.

TS-LBS test system by Rohde & Schwarz (R&S), which consists of CMX500 and CMW500 radio communication testers and SMBV100B vector signal generator, covers 3rd Generation Partnership Project (3GPP) conformance and network operator test plans for location-based services of 5G NR, LTE, WLAN, WCDMA and GSM.

To calibrate OTA test setups and to carry out OTA measurements, R&S offers a diverse portfolio of vector network analysers, signal and spectrum analysers, and OTA test solutions. Antenna test chambers such as ATS1000 are used for 5G antenna characterisation throughout the process, from R&D to production for both active and passive devices.

Protocol conformance testing. This enables mobile operators, chipset and device manufacturers, as well as test labs to speed validation of new devices, from early design to acceptance and manufacturing. This allows mobile operators to deploy 5G NR in